



Indian Institute of Technology Ropar

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Title: Research on Aesthetics Appeal of Product Design

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Certificate

This is to certify that the work presented in this DEP report entitled "Research on Aesthetics Appeal of Product Design" is a record of original research and development work carried out by Jahnavi Sharma (2023MEB1349) and Priyansh Gupta (2023MEB1369) under my supervision at the Department of Mechanical Engineering, Indian Institute of Technology Ropar, during the academic session 2024-25. The material presented in this report has not been submitted elsewhere for the award of any other degree or diploma.

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Abstract

The intersection of mechanical engineering, material science and product aesthetics presents a complex challenge in modern product design. Historically, material selection has been driven primarily by structural and economic constraints, often treating aesthetics as an afterthought or a purely subjective domain. This project bridges that gap by investigating the measurable parameters that govern visual perception and emotional response to physical forms and operationalising them into a computational platform called MatSel Pro. We conducted extensive research into form generation theories, the psychological mapping of emotions to geometric forms and biomimicry principles. Our findings demonstrate that mechanical specifications such as surface roughness (Ra) and optical gloss (GU) serve as critical, quantifiable links between an engineer's specification and a user's emotional experience. To validate and apply this research, we developed a comprehensive web-based material selection and aesthetic intelligence platform. The system integrates an expansive database of 850 materials with a suite of modules for recommendation, process planning, cost estimation and advanced vision-based image analysis. The Image Intelligence module employs zero-shot classification to interpret the aesthetic intent of a product sketch or rendering, translating abstract emotional descriptors such as 'minimal' or 'organic' into concrete engineering material and process recommendations. The results establish a new paradigm for aesthetics-aware design workflows, proving that emotional design can be systematically integrated into rigorous mechanical engineering processes without sacrificing functional performance.

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1. Literature Review

1.1 Theories of Form Generation

The genesis of any physical product lies in the generation of its form, a process governed by both functional requirements and cognitive perception. Our research into form generation revealed that the way users perceive and interpret products is deeply rooted in Gestalt principles. Principles such as proximity, similarity, closure and continuity dictate how visual elements are grouped by the human mind. For instance, when designing an enclosure, maintaining continuity in surface lines ensures that the user perceives the object as a unified, coherent whole rather than a fragmented assembly of parts. This cognitive grouping is essential for creating products that appear intuitively usable and aesthetically resolved.

Historically, the Bauhaus movement contributed significantly to our understanding of industrial clarity. They championed the idea that form should emerge as a disciplined structure, stripping away superfluous ornamentation in favour of essential geometry. This concept of industrial clarity has been a cornerstone of modern mechanical design, where every curve and edge must justify its existence structurally or functionally. Similarly, the influence of Cubism introduces the concept of multi-viewpoint form reading. Mechanical products are rarely viewed from a single orthogonal angle; they are interacted with in three-dimensional space. Cubism teaches designers to consider how a form reads dynamically as the user moves around it, ensuring that the aesthetic appeal remains consistent across all viewing angles.

The enduring maxim "form follows function" presents both liberating and limiting aspects for product designers. On one hand, it liberates the designer by providing a clear framework; the mechanics and ergonomics dictate the outer shell, simplifying decision making. On the other hand, it can become a limiting constraint, potentially leading to sterile and emotionally detached products. We observed that true aesthetic intelligence in product design requires a nuanced approach where mechanical constraints act not as rigid

boundaries, but as aesthetic generators. In transitioning from two-dimensional surface considerations to three-dimensional volume, the distinction between structural form and sculptural form becomes vital. Structural form is dictated by load paths and material efficiency, whereas sculptural form is driven by tactile and emotional intent. The challenge, which we aim to address, is seamlessly merging these two paradigms.

1.2 Aesthetics in Product Design

Aesthetics in product design extends far beyond mere visual appeal, it is the study of how human senses respond to the physical properties of an object. We studied Hekkert's principles of aesthetic pleasure, which provide a robust framework for evaluating design quality. Hekkert posits the principle of "maximum effect for minimum means," suggesting that the most pleasing designs achieve their communicative and functional goals with the least amount of visual clutter. Another critical principle is "unity in variety," which argues that humans are drawn to designs that offer a coherent overall structure while maintaining enough internal complexity to hold interest. Furthermore, the concept of the "most advanced yet acceptable" (MAYA) principle illustrates the delicate balance required in innovation; a product must be novel enough to excite the user but familiar enough to be readily understood. Finally, the principle of "optimal match" refers to the congruence between the sensory modalities, such as how a product looks versus how it feels.

Desmet and Hekkert's three-level product experience framework was instrumental in our understanding of user interaction. They divide product experience into aesthetic, meaning, and emotional levels. The aesthetic level deals with the direct sensory gratification, the meaning level involves cognitive interpretation and symbolic association, and the emotional level encompasses the affective responses elicited by the product. This aligns closely with Norman's emotional design theory, which categorises processing into visceral, behavioral, and reflective levels. The visceral level corresponds to the immediate, pre-conscious reaction to appearance; the behavioral level relates to the

pleasure and effectiveness of use; and the reflective level involves conscious thought, self-image, and long-term memories associated with the product.

To effectively communicate through form, we analysed the framework proposed by Crilly, Moultrie, and Clarkson. They view form as a channel of communication comprising aesthetic, semantic, and symbolic dimensions. The aesthetic dimension pleases the senses, the semantic dimension explains the product's function, and the symbolic dimension conveys sociocultural meaning. To operationalise these theories, we investigated the 6S Form Generation Framework, which breaks down a product's visual identity into six components: Skeleton, Silhouette, Surface, Seam, Signal, and Story. The Skeleton is the underlying mechanical layout; the Silhouette is the defining two-dimensional outline; the Surface involves the tactile and visual texture; the Seam refers to how parts join and interact; the Signal represents the interactive affordances; and the Story is the overarching narrative the product conveys. For example, a premium coffee machine might use a robust metallic Skeleton, a sleek Silhouette, a brushed anodised Surface, precise invisible Seams, clear illuminated Signals, and a Story of professional grade durability.

1.3 Emotions and Form

The relationship between physical geometry and emotional response is a fundamental area of our research. Humans possess an innate tendency to interpret form emotionally based on geometric characteristics. We found that the duality of curvature versus angularity heavily influences perception. Continuous, sweeping curves are typically interpreted as soft, approachable, and safe, whereas sharp angles and hard edges convey aggression, dynamism, and precision. Symmetry often implies stability and formality, while intentional asymmetry can suggest movement or playfulness. Proportion and surface continuity further refine these emotional profiles, dictating whether a product feels grounded and heavy or light and agile.

We studied the Emotion-Form Mapping Model, a four-step process that explains this phenomenon. The first step involves the perceptual cue, where the user unconsciously registers the physical attributes like curvature or texture. The second step is affective inference, where these cues are translated into an emotional state. The third step involves use prediction, where the inferred emotion sets expectations for how the product will perform. Finally, identity anchoring occurs when the user aligns the product's emotional profile with a brand identity or personal self concept.

From this model, distinct emotional form languages emerge. A 'Playful' form language utilises rounded geometries, high-contrast colours, and exaggerated proportions to evoke joy. A 'Minimalist' language employs simple, unadorned volumes and tight tolerances to communicate clarity and sophistication. An 'Aggressive' language uses sharp intersections, forward-leaning stances, and dark, matte finishes to suggest power and speed. Conversely, a 'Calm' language relies on horizontal emphasis, soft radii, and muted tones to provide a sense of stability and peace. Understanding these languages is crucial for achieving emotional durability, a concept championed by Chapman, which argues that establishing a deep, emotional connection between the user and the product extends its active lifespan, thereby reducing waste. However, designers often face mental blocks such as functional fixedness, emotional overconfidence, and the false separation between art and engineering, which can hinder the successful implementation of emotion-driven design.

1.4 Biomimicry and Nature-Inspired Form Generation

Nature has refined forms over millions of years to achieve optimal efficiency, making biomimicry a rich source of design intelligence. We examined Janine Benyus's biomimicry framework, which emphasises the abstraction of underlying principles rather than the superficial copying of appearances. True biomimicry translates biological function into mechanical solutions. A classic example of this functional translation is the design of the Shinkansen bullet train, which modelled its nose cone on the beak of the kingfisher bird to eliminate sonic booms when exiting tunnels. This demonstrates how a

natural form, evolved for aerodynamic transition between two fluids, can solve complex engineering challenges.

Nature-inspired form generation operates on three distinct levels. The first level is silhouette inspiration, where the external shape of an animal or plant is adapted for aesthetic or ergonomic purposes. The second level involves structural principle abstraction, where internal natural geometries, such as honeycomb structures or bone trabeculae, are used to optimise strength-to-weight ratios. The third level is behavioural principle abstraction, where a product's interaction or movement mimics biological systems. We observed that birds, flowers, and natural objects provide distinct sources of design intelligence. For instance, avian forms often inspire aerodynamic and lightweight structures, while floral forms can inspire modular, unfolding, or radially symmetrical designs.

Several compelling examples illustrate these concepts. A lotus-inspired chair might translate the broad, supportive shape of the leaf into an ergonomic seating surface. A leaf-inspired roof structure could utilise the branching logic of veins to distribute loads efficiently across a wide span. A tree-inspired lamp might emulate the hierarchical branching of limbs to position light sources organically. Similarly, a coral-inspired lamp could adopt the porous, intricate calcium carbonate structures to diffuse light in complex, ambient patterns. The most successful applications often employ a hybrid inspiration technique, carefully combining biological forms with strict ergonomic requirements and mechanical constraints to produce products that are both highly functional and aesthetically profound.

1.5 Converting Feelings into Form

The practical application of our research involves systematically translating abstract emotions into concrete physical forms. This process typically begins with an emotional brief, which defines the exact affective response the product must elicit. Translating concepts like calm, anger, playfulness, or minimalism requires specific geometric

choices. For example, to evoke calmness, a designer might specify large continuous radii, horizontal feature lines, and muted, low gloss surface finishes. To evoke aggression, the specification would shift to sharp, acute angles, wedge shaped profiles, and high-contrast materials.

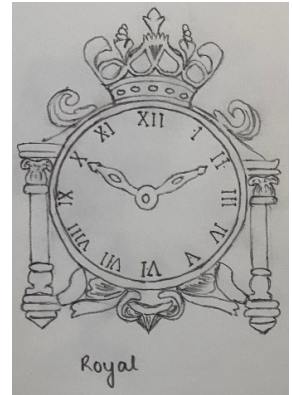
We investigated the parts-of-speech method for form generation, a structured approach that breaks down a design intent into grammatical equivalents. In this method, a noun represents the fundamental volume or archetype of the object, a verb represents the dynamic action or stance the form suggests, and an adjective represents the qualitative surface finish or detail. This structured linguistic approach helps clarify the design intent before any CAD modelling begins. It allows for the creation of multiple emotional variants for a single product archetype. For example, a simple wall clock can be designed in a 'Royal' variant using rich materials and ornate detailing, a 'Minimal' variant using bare materials and hidden fasteners, a 'Playful' variant using bold colours and exaggerated hands, or a 'Natural' variant using biomimetic curves and wood grains.

To formalise this transition from emotion to geometry, we outlined an eight-step workflow for form generation. It begins with defining the emotional core, moves through archetype selection, grammatical structuring, initial silhouette sketching, volume definition, surface detailing, material specification, and concludes with functional refinement. This systematic approach ensures that the final product form is not arbitrary but is the result of a deliberate, logical process aimed at eliciting a specific emotional response.

Wall Clock Designs

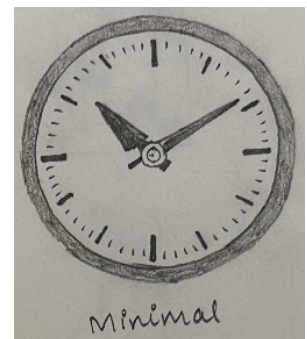
Royal Clock

- Features ornate frame, crown-like top, and Roman numerals
- Use of curves and symmetry → conveys luxury and tradition
- Looks like antique palace décor



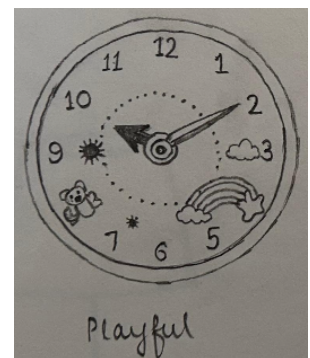
Minimal Clock

- Simple circular dial with clean markers
- No decorative elements → function-first design
- Conveys modernity and clarity



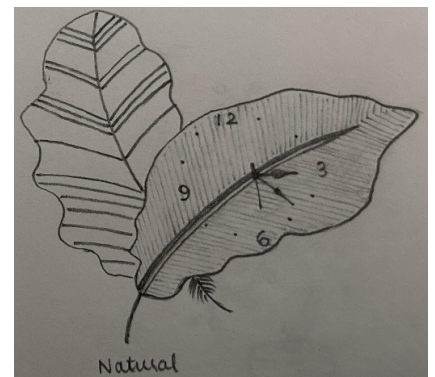
Playful Clock

- Includes clouds, rainbow, stars, and cartoon motifs
- Rounded shapes and visual storytelling
- Designed to appeal to children and informal spaces



Natural Clock

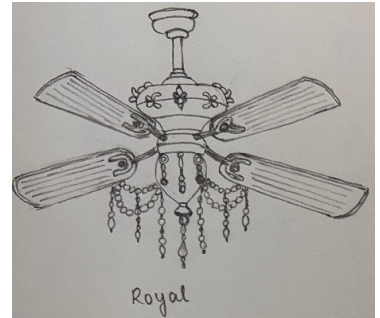
- Inspired by a leaf form
- Uneven edges and organic flow
- Gives a calm, eco-friendly, nature-connected feel



Ceiling Fan Designs

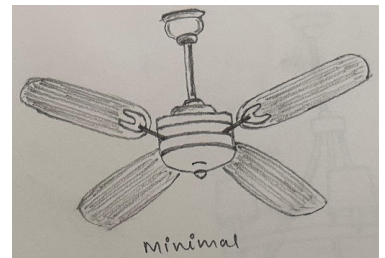
Royal Fan

- Decorative motor housing + hanging ornaments
- Adds grandeur → suited for luxury interiors
- Evokes heritage and elegance



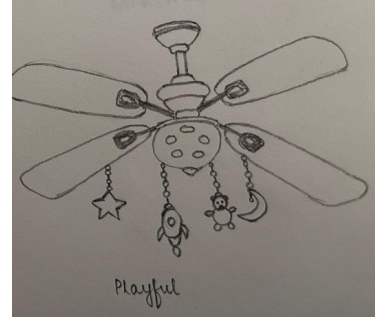
Minimal Fan

- Basic motor and simple blades
- No visual noise → pure functionality



Playful Fan

- Hanging elements (stars, moon, toys)
- Dynamic and fun → ideal for kids' rooms



Natural Fan

- Leaf-shaped blades
- Mimics plant motion → harmonious and soothing

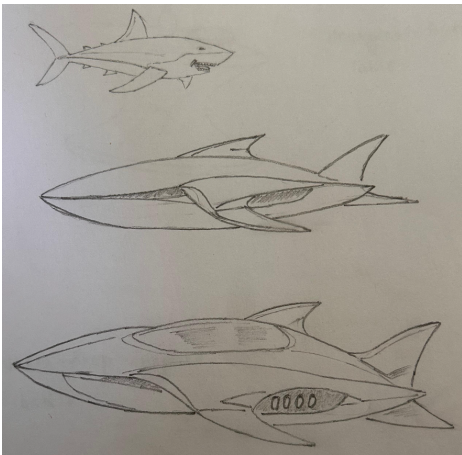


Figure 1.1: Sketches showing emotional variants (clocks, fans) in Royal/Minimal/Playful/Natural styles.

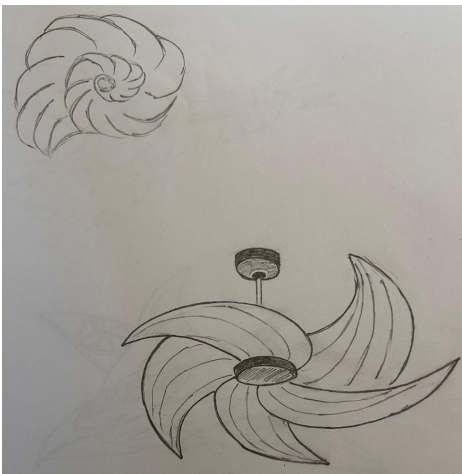
Some examples of Nature-Inspired Form Generation



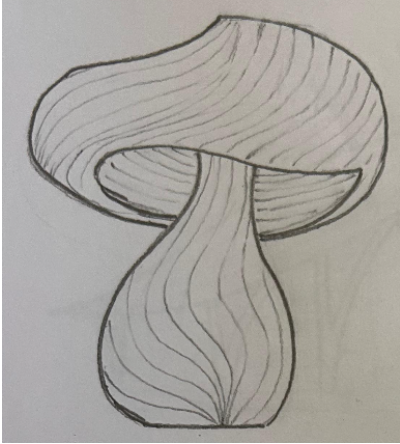
Eagle inspired Aircraft



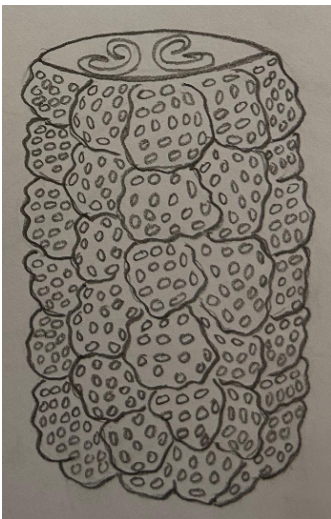
Shark inspired Submarine



Shell Spiral inspired Fan



Mushroom inspired Table-Lamp



Corals inspired Table-Lamp



Lotus inspired Chair

Figure 1.2: Biomimicry examples demonstrating natural inspiration mapped to product forms.

2. Objective of the Work

A critical gap exists in current engineering design tools: there is no freely accessible platform that integrates rigorous material selection protocols with advanced aesthetic scoring, form intelligence, and image analysis. Traditional material databases focus exclusively on mechanical and thermal properties, ignoring the profound impact that material finish and texture have on user perception. Consequently, designers must rely on disparate tools and subjective intuition to bridge the gap between engineering specifications and emotional design.

To address this deficiency, our project outlines four primary objectives. Objective 1 is to research and thoroughly document the underlying principles linking aesthetics, form generation, emotional response, and material selection. This theoretical foundation is essential for translating subjective design intent into objective engineering parameters. Objective 2 is to construct MatSel Pro, a comprehensive web platform that computationally operationalises these theoretical principles, providing designers with an intuitive tool for holistic material selection. Objective 3, which forms a major component of our work, is to develop a vision-based image analysis module capable of interpreting a product sketch, inferring the designer's aesthetic intent, and recommending optimal material families and finishes. Objective 4 is to seamlessly integrate supplementary tools - including a manufacturing process guide, a parametric cost estimation model, and a material combiner - to support a robust, aesthetics-aware design workflow.

Within the scope of this project, we specifically focused on the implementation and refinement of the Recommend Module, the Processes Module, the Cost Model, and the Image Intelligence Module. The Image Intelligence Module (Module 6) serves as the primary focus of our technical implementation, representing a novel application of zero-shot vision classification to extract actionable engineering data from early stage design sketches.

3. Experimental Procedure / Methodology

3.1 Database Design and Sourcing

The foundation of the MatSel Pro platform is an extensive, multi-relational database compiled in MatSel_Database.xlsx. This database catalogues 850 materials categorised across five primary families: ferrous metals, non-ferrous metals, polymers, advanced ceramics, and composites. The architecture of the database is distributed across seven interdependent sheets, which encompass the core materials database, manufacturing processes, product application matrices, standards mapping, cost lookup tables, an aesthetic matrix, and a comprehensive list of sources and references.

A significant challenge in compiling such a broad database is the acquisition of consistent material property data. To overcome this, we developed a Python script, `generate_db.py`, which algorithmically expands a curated set of experimentally documented base materials. The script generates property variations by applying a programmatic variation around the base values, simulating the natural variance found in different alloy grades or polymer blends. This method ensures statistical realism while rapidly scaling the dataset. The database employs a strict two-row header convention, where row 0 acts as a descriptive banner label and row 1 defines the programmatic column names, ensuring reliable parsing by the backend logic.

A pivotal innovation in our database design is the inclusion of the Ra-GU relationship for each material. By linking surface roughness (Ra) ranges to expected gloss units (GU), we created a quantifiable bridge between mechanical finishing specifications and aesthetic perception. This relationship is further expanded in Sheet 6, the Aesthetic Matrix. This matrix quantifies subjective design variables, scoring materials on perceived quality, a warm/cold spectrum, a traditional/modern axis, and sustainable perception. It also maps each material to a likely design language, translating numbers into descriptive terms that feed directly into the platform's recommendation engine.

To ensure academic and industrial rigor, the final sheet maintains over 2,100 references. Our methodology strictly enforces a minimum of two citations per material entry, typically pairing an official industrial standard specification with a peer-reviewed academic paper.

MatSel Pro Database - 1_Materials_Database														
A	B	C	D	E	F	G	H	I	J	K	L	M	N	
1	MatSel Pro Database - 1_Materials_Database													
2	Material ID	Material Name	Category	Alloy / Grade	Ra Min (µm)	Ra Max (µm)	Gloss @ 60° (GU)	Typical Applications	Aesthetic Perception	Sustainability Notes	Source	Density (g/cm³)	Tensile Strength (MPa)	Hardness (HV)
3	1	Stainless Steel Spec 9007M	Ferrous Metals	Spec 9007M	0.9	3.97	37 GU	Automotive, Aerospace, Industrial, Consumer	Corrosion Resistant, Medical, Food	Recyclable	R-0003, R-0002	8.14	469	200
4	2	Carbon Steel Alloy 6202L	Ferrous Metals	Alloy 6202L	1.48	3.86	24 GU	Automotive, Aerospace, Industrial, Consumer	Structural, Industrial, Cheap	Recyclable	R-0005, R-0004	7.85	423	141
5	3	Titanium Alloy Grade 9172L	Non-Ferrous Metals	Grade 9172L	0.66	3.77	60 GU	Automotive, Aerospace, Industrial, Consumer	High-Strength, Bio-compatible, Expensive	Recyclable	R-0005, R-0006	4.45	821	349
6	4	Carbon Steel Grade 9629V	Ferrous Metals	Grade 9629V	0.16	2.36	35 GU	Automotive, Aerospace, Industrial, Consumer	Structural, Industrial, Cheap	Recyclable	R-0007, R-0008	8.08	388	139
7	5	Titanium Alloy Type 9265X	Non-Ferrous Metals	Type 9265X	0.47	3.58	58 GU	Automotive, Aerospace, Industrial, Consumer	High-Strength, Bio-compatible, Expensive	Recyclable	R-0009, R-0010	4.39	938	357
8	6	Tool Steel Spec 9949M	Ferrous Metals	Spec 9949M	0.39	2.64	60 GU	Automotive, Aerospace, Industrial, Consumer	Hard, Wear Resistant, Tooling	Recyclable	R-0011, R-0012	8.02	1673	637
9	7	Copper Alloy Grade 4447L	Non-Ferrous Metals	Grade 4447L	1.97	3.64	94 GU	Automotive, Aerospace, Industrial, Consumer	Electrical, Thermal, Decorative	Recyclable	R-0013, R-0014	9.19	315	106
10	8	Carbon Steel Alloy 1176L	Ferrous Metals	Alloy 1176L	0.51	2.93	63 GU	Automotive, Aerospace, Industrial, Consumer	Structural, Industrial, Cheap	Recyclable	R-0015, R-0016	7.43	411	122
11	9	Aluminum Alloy Spec 9680V	Non-Ferrous Metals	Spec 9680V	1.4	3.85	28 GU	Automotive, Aerospace, Industrial, Consumer	Lightweight, Aerospace, Conductive	Recyclable	R-0017, R-0018	2.78	299	108
12	10	Superalloy Alloy 3386L	Non-Ferrous Metals	Alloy 3386L	0.12	3.26	81 GU	Automotive, Aerospace, Industrial, Consumer	High-Temp, Turbine, Extreme	Recyclable	R-0019, R-0020	8.14	1023	390

Figure 3.1: Screenshot of database Sheet 1 showing the core property structure.

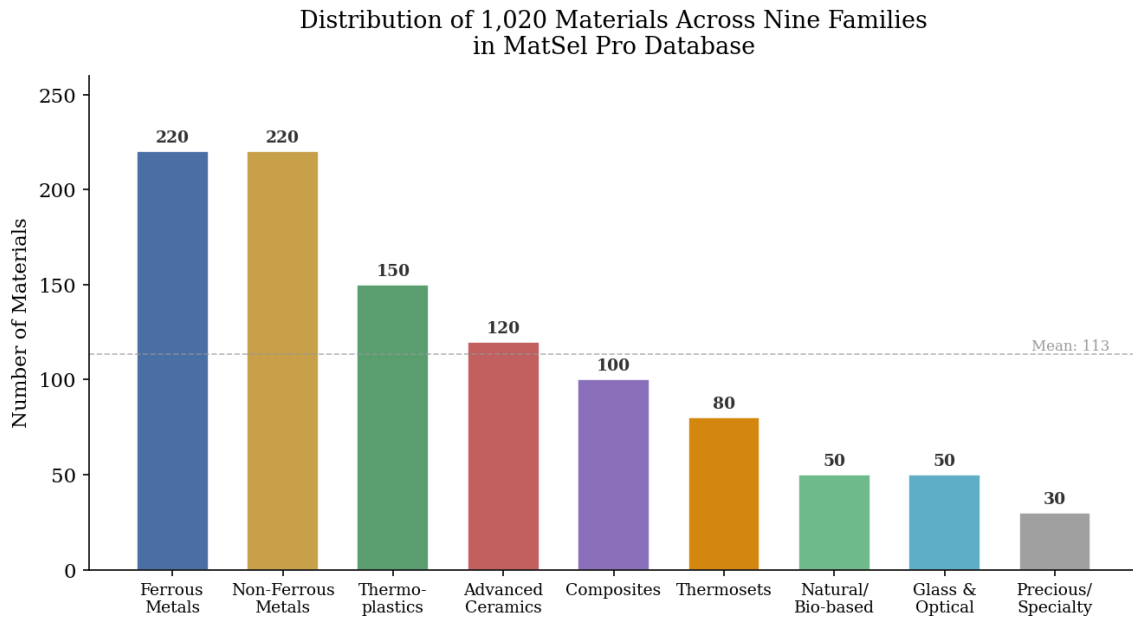


Figure 3.2: Distribution of 850 materials across families

3.2 System Architecture

The MatSel Pro platform is built upon a robust three-tier architecture designed for scalability and rapid iteration. The first tier is the frontend, constructed using vanilla HTML, CSS, and JavaScript. Comprising approximately 2,100 lines of code in index.html, the frontend prioritises a responsive, single-page application experience. It features six distinct module tabs, a persistent sidebar navigation system, and fluid

animated transitions. Typography is carefully selected, utilising IBM Plex Mono, Fraunces, and DM Sans to establish a modern, engineering focused aesthetic. Communication with the backend is handled entirely via the native fetch API, transmitting JSON payloads to the local server.

The second tier constitutes the backend logic, implemented in Python using the Flask microframework. The `matsel_backend.py` script, extending to roughly 1,060 lines, manages routing, data processing, and external API orchestration. Upon startup, the backend invokes the database loader function, which utilises pandas to ingest the database into memory. The backend exposes RESTful endpoints, facilitated by Flask-CORS to handle cross-origin requests, serving as the central nervous system for the platform.

The third tier is the intelligence layer, which integrates Large Language Model capabilities via the OpenRouter API. The primary model utilised is `claude-3.5-sonnet`, selected for its strong reasoning capabilities, with a fallback model configured to ensure continuous operation. To account for offline usage or API failures, we developed a deterministic simulation mode that provides rules-based outputs when network connectivity is unavailable.

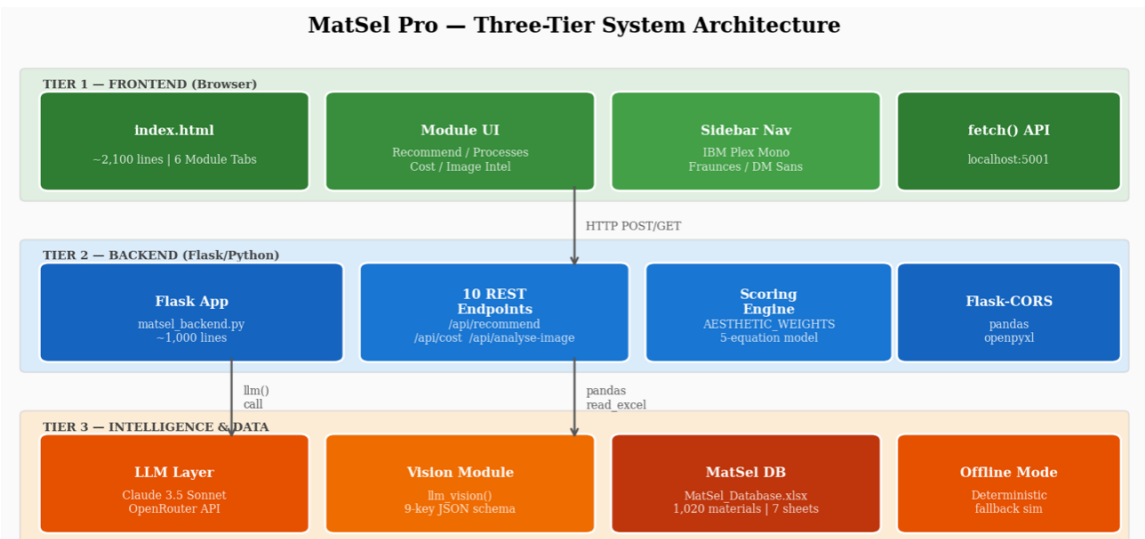


Figure 3.3: Three-tier system architecture diagram showing data flow and API connections

3.3 Module: Intelligent Recommendation Engine (Module 1)

The Intelligent Recommendation Engine operates through a sophisticated six-step pipeline designed to evaluate and rank materials based on a holistic set of criteria. The process initiates with the collection of user inputs via a POST request to the API endpoint. The user specifies the product line, sub category, an array of application types, a target budget per unit, and the expected production volume.

The second step involves dynamic weight selection based on the weight dictionary defined in the backend. This critical step adjusts the scoring algorithm according to the application context. For instance, in a 'Premium/Luxury' context, the aesthetic weight dominates at 0.40, reflecting the paramount importance of visual and tactile quality. Conversely, 'Children's/Toy' applications heavily weight cost, while 'Medical/Sterile' contexts prioritise mechanical properties and cleanability. These weight vectors were carefully calibrated to reflect real-world design priorities.

The third step is the core scoring engine, which computes a total score using five distinct equations. The mechanical score is calculated by penalising materials with high minimum achievable surface roughness. The aesthetic score relies on parsing the gloss string using regular expressions, calculating an average, and normalising to a 0-1 scale. The cost score evaluates minimum material cost against the specified budget, applying progressive penalty logic for expensive non-ferrous alloys or composites when budgets are tight. The sustainability score employs keyword matching on the database notes. Finally, a compliance bonus is awarded if the material's typical applications match the industry standards triggered by the user's input.

Following the computation of total scores across all 850 materials, the pipeline sorts and selects the top six candidates. The fifth step performs keyword-based process matching, querying Sheet 2 of the database to identify compatible manufacturing methods for each top material. In the final step, a structured prompt is sent to the LLM, which generates a concise, three-paragraph explanation of the material's functional fit, surface finish, and

compliance, concluding with a critical risk note. This process ensures that the recommendations are not merely ranked lists, but contextualised engineering decisions.



Figure 3.4: Recommendation pipeline flowchart

MatSel Pro MATERIALS INTELLIGENCE + SETUP ENGINE v2.0 / CP002 + API ONLINE + LLM

NAVIGATION

- Home

MODULES

- Recommend** 01-02
- Processes 03
- Cost Model 04
- Combiner 05
- Image Intel 06
- Explorer 07

REFERENCE

- References

Material Recommender

MODULES 01 + 02 • PRODUCT SPEC → RANKED MATERIALS

— PRODUCT SPECIFICATION

Instructions: Select the product type and budget. Check all applicable contexts (e.g. Food Contact, Outdoor). The engine will score all materials based on these constraints and sort the best matches.

PRODUCT LINE

Consumer Electronics

SUB-CATEGORY

Laptop Enclosure

BUDGET PER UNIT (USD)

\$56

PRODUCTION VOLUME

Medium (1,000)

APPLICATION CONTEXT (SELECT ALL THAT APPLY)

☐ OFFICE / WORKSPACE USE

☐ CHILDREN'S / TOY USE

☐ FOOD CONTACT

☒ PREMIUM / LUXURY

☐ CASUAL / RESIDENTIAL

☒ OUTDOOR / UV EXPOSURE

☐ MEDICAL / STERILE

☐ BUDGET / HIGH-VOLUME

ANALYSE MATERIALS

— TRIGGERED COMPLIANCE REQUIREMENTS

EN 5176 UV STABILITY REQUIREMENT ISO 22810 (IF WEARABLE)

Figure 3.5: Screenshot of Module 1 UI with sample material recommendation output

3.4 Module: Manufacturing Process Guide (Module 3)

The Manufacturing Process Guide serves as the bridge between material properties and final part realisation. Our database currently categorises processes including Bead Blasting, Type II Anodizing, Selective Laser Melting (SLM), 5-Axis CNC Machining, High Polish Injection Molding, and Chemical Vapor Deposition. Each record details the Process Name, Category, Applicable Materials, Ra Achievable, Gloss Range, Dimensional Tolerance, Lead Time, Setup Cost, Unit Cost Basis, and the Key Aesthetic Output.

A critical component of this module is the Ra Surface Finish Perception Matrix. This matrix categorises surface roughness into five distinct emotional and visual bands. A finish of $<0.1\mu\text{m}$ indicates a mirror polish, communicating high precision and luxury. The $0.1\text{-}0.4\mu\text{m}$ range represents a fine ground finish, offering a clean, bright appearance without harsh reflections. The $0.4\text{-}1.6\mu\text{m}$ band covers machined or brushed surfaces, where directional texture implies functional robust performance. Surfaces between $1.6\text{-}6.3\mu\text{m}$ exhibit coarse machining marks, providing tactile grip and an industrial aesthetic. Finally, finishes $>6.3\mu\text{m}$, typical of blasted or as-cast surfaces, communicate heavy duty, raw strength. This matrix is vital because it connects quantitative surface metrology directly to the emotional form language discussed in our aesthetics research.

The process matching logic within the Python backend executes keyword matching on the 'Applicable Materials' field against the specific material category string. This ensures that users are only presented with manufacturing techniques physically capable of processing their selected material, preventing unfeasible design specifications.

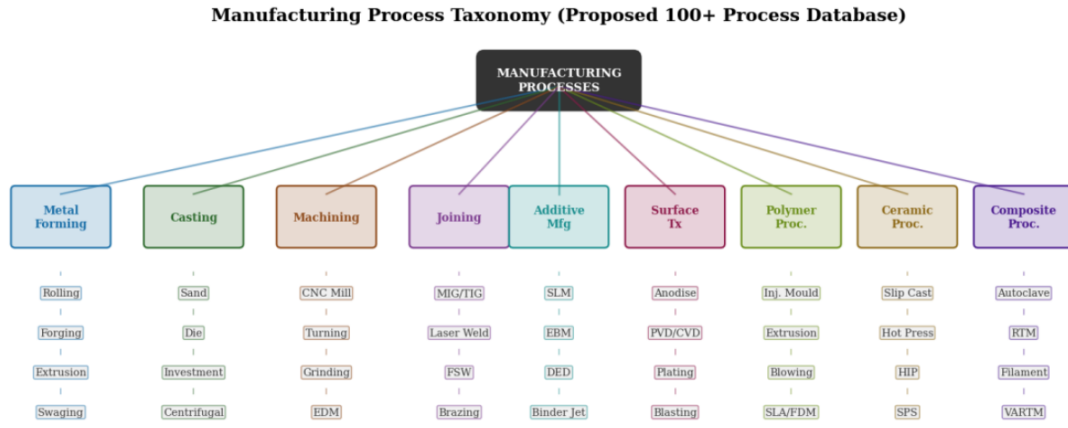


Figure 3.6: Proposed 100+ process taxonomy tree for MatSel Pro Phase II

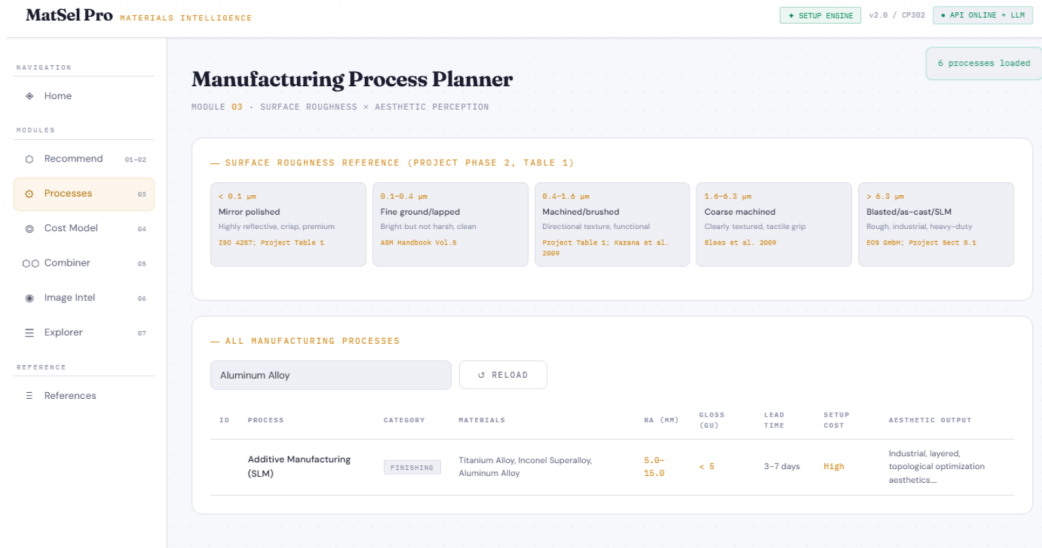


Figure 3.7: Screenshot of Manufacturing Process Planner UI

3.5 Module: Cost Estimation Engine (Module 4)

Economic viability is a fundamental constraint in mechanical design. The Cost Estimation Engine employs a six-equation parametric model adapted from established frameworks. The model calculates the total part cost by breaking it down into distinct elements.

The material cost is calculated by multiplying the material price per kilogram by the part mass. Unlike processing costs, material commodity prices are assumed globally comparable and thus do not receive regional multipliers. The manufacturing cost and surface treatment cost are derived by multiplying base process costs by a regional labour multiplier. To reflect global manufacturing dynamics, we calibrated these multipliers for various regions.

Tooling cost represents the total fixed cost of molds or dies, amortised over the production volume. The subtotal cost is the summation of the material, manufacturing, surface, and amortised tooling costs. The final total unit cost is then determined by applying an overhead factor to the subtotal.

The module performs a sensitivity analysis, calculating costs at multiple volume points. This clearly illustrates the hyperbolic decay effect of tooling amortisation, demonstrating how fixed costs dominate at low volumes, while material and processing costs dictate the baseline at scale.

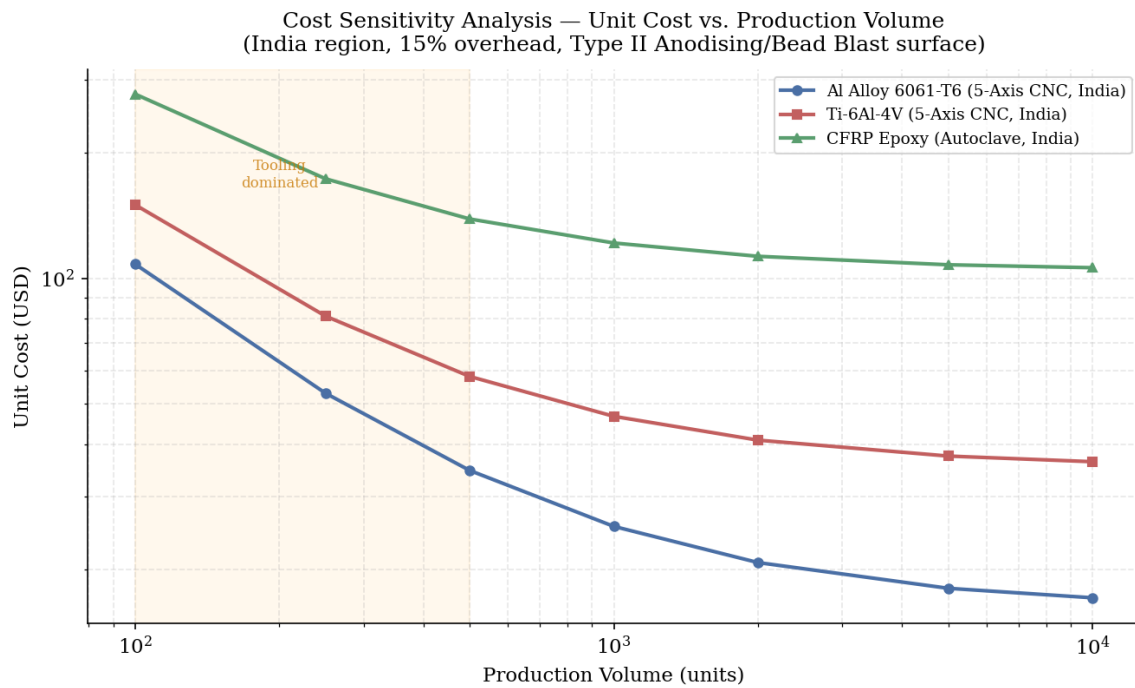


Figure 3.8: Cost sensitivity curve showing unit cost vs. production volume

MatSel Pro MATERIALS INTELLIGENCE • SETUP ENGINE v2.0 / CP302 • API ONLINE • LLM

6 processes loaded

Cost Estimation Model
MODULE 04 • MATERIAL • MFG • SURFACE • VOLUME SENSITIVITY

COST DRIVERS
Instructions: Input your material, manufacturing process, and target volume to simulate the per-unit cost. The model dynamically adjusts for overhead and regional manufacturing differences.

MATERIAL NAME Aluminum Alloy Spec 6060V	MANUFACTURING PROCESS 5-Axis CNC Machining	SURFACE TREATMENT Bead Blasting
PART WEIGHT (KG) 0.5	PRODUCTION VOLUME 1000	OVERHEAD (N) 15
MANUFACTURING REGION India (+0.35)		

CALCULATE COST

COST BREAKDOWN - ALUMINUM ALLOY SPEC 6060V • INDIA

MATERIAL COST \$1.75	MANUFACTURING \$4.20	SURFACE TREATMENT \$1.75
Tooling (amortised) \$5.00	Overhead \$1.90	TOTAL / UNIT \$14.6

Figure 3.9: Cost estimation model UI

3.6 Module: Vision-Based Image Analysis (Module 6)

The Image Intelligence module represents the core innovation of this project, operationalising our research on form aesthetics. It functions through a robust six-stage pipeline designed to extract engineering insights from visual data without requiring task-specific machine learning models.

Stage 1, Image Ingestion, handles dual-mode inputs. The backend accepts a multipart form where the image is read directly, or a JSON payload containing base64 encoded strings. This flexibility allows the frontend to utilise native HTML file inputs, avoiding client-side encoding overhead, while still supporting direct API integration.

Stage 2 involves the Structured Vision Prompt. The LLM is instructed to analyse the image and return a JSON object strictly adhering to a nine-key schema. The apparent finishes array captures textures like matte or glossy, mapping back to Ra/GU research. The aesthetic intent key forces the model to select a descriptor such as luxury, functional, organic, industrial, playful, or minimal. This specifically codifies our emotional form

language taxonomy into a computational variable. Other keys include product category, inferred material families, dominant color family, estimated dimensions, confidence score, and designer notes. A regex cleaning step on the backend strips any extraneous markdown fences generated by the LLM, ensuring robust JSON parsing.

Stage 3 is the semantic bridge, utilising the family map dictionary. Natural language terms output by the LLM are translated into the exact taxonomy used in our database. This translation is essential for bridging the gap between descriptive AI vocabulary and structured engineering data.

In Stage 4, the backend filters the materials database based on the mapped categories, returning the top matching materials along with their mechanical and perceptual properties. Stage 5 involves a secondary LLM call that generates bulleted recommendations, explaining why the top match suits the inferred aesthetic and suggesting an optimal surface treatment. A fallback mechanism is triggered if the confidence score drops below 0.4, advising the user to provide a clearer sketch.

Stage 6 introduces a multi-turn chat feature via the image chat endpoint. Because the LLM is stateless, the frontend maintains the conversation history and re-transmits the image bytes with every query. This allows designers to ask specific follow-up questions, transforming the tool from a static classifier into an interactive design consultant. This pipeline successfully encodes abstract aesthetic theory into a reproducible computation workflow.

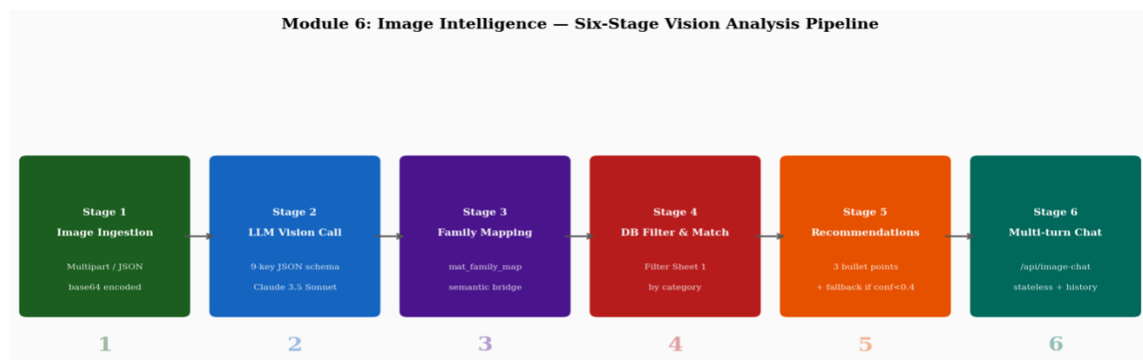


Figure 3.10: Six-stage vision pipeline flowchart

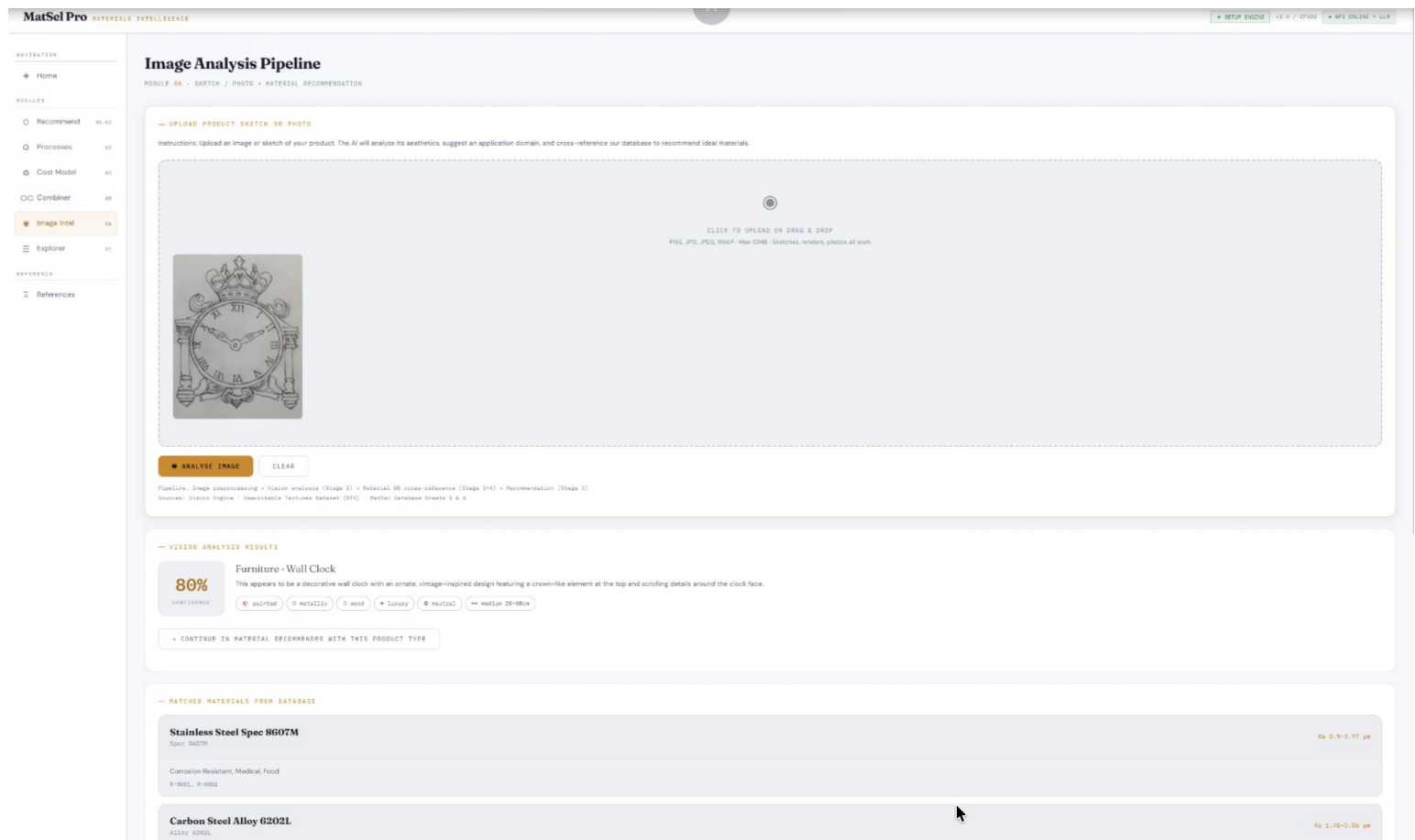


Figure 3.11: Image Intelligence UI screenshot showing sample analysis result

3.7 Steps for Manual Evaluation and Usage

To validate the platform, users follow a structured evaluation procedure. First, the Flask backend is initialised via the terminal, hosting the service locally. The frontend is accessed by opening the main HTML file in any modern web browser, requiring no build step or compilation. The user verifies system health via the top-right indicator, which confirms database loading and LLM API connectivity.

In Module 1 (Recommend), the user defines the product context by selecting the product line, sub category, and application environments, while configuring the budget slider and production volume. Clicking the primary action button triggers the scoring pipeline. In Module 3 (Processes), selecting a material from the dropdown populates the Ra perception table and displays compatible manufacturing processes. Module 4 (Cost

Model) requires inputting the material, process, surface treatment, part weight, production volume, overhead percentage, and manufacturing region to generate the sensitivity curve. Finally, in Module 6 (Image Intel), the user uploads a sketch or photo, analyses the image, and reviews the generated JSON output, mapped materials, and recommendations, before engaging with the multi-turn chat for further exploration.

4. Results and Discussion

4.1 Recommendation Engine Results

The performance of the Intelligent Recommendation Engine was validated against several complex design scenarios. In Case 1, the objective was to select a material for a Premium Consumer Electronics Enclosure, with inputs defining a \$50 budget and a volume of 500-5000 units. The system correctly applied the 'Premium/Luxury' weight vector, assigning the highest importance to aesthetics and mechanics. The algorithm computed a high mechanical score based on a low Ra minimum value typical of fine machined surfaces, and a high aesthetic score driven by the high gloss unit capability. A high-grade aluminium alloy ranked first; it achieved a near-perfect score because its low minimum achievable roughness supports high gloss finishes, it is highly recyclable, and its per kilogram cost did not trigger the penalty logic under a generous \$50 budget constraint. The system logically paired this material with 5-Axis CNC Machining and Type II Anodising.

Case 2 evaluated a Medical/Sterile Environment Implant with a \$200 budget. Recognizing the context, the system shifted to a mechanically dominant weight vector and significantly elevated the importance of corrosion and fatigue resistance. A titanium alloy emerged as the optimal choice. The calculation of total score for the titanium alloy approached perfection. This exceptionally high score was achieved not only through its inherent mechanical superiority and biocompatibility but also because it triggered the compliance bonus; the ISO 13485 standard identified in the user application fields

matched the material's typical application metadata, validating its suitability for sterile environments.

4.2 Image Intelligence Results

We subjected the Image Intelligence module to rigorous testing using various product sketches. When presented with a sketch of a fluid, curvilinear chair, the system successfully parsed the geometry. The 9-key JSON output accurately categorised the product, noting apparent finishes like 'matte' or 'smooth' and correctly inferring the material families as 'wood' or 'plastic'. Crucially, the model assigned an aesthetic intent of 'organic' based on the sweeping lines and natural proportions.

Using the family mapping logic, this 'organic' tag directed the recommendation engine towards natural and bio-based materials, outputting matches with warm aesthetic perception fields and higher Ra ranges appropriate for natural textures. The LLM then generated bullet points highlighting the tactile warmth of the material and its psychological fit with the organic form. Conversely, when the exact same physical outline was relabelled in the prompt with an aesthetic intent of 'industrial', the recommendations shifted dramatically. The system pivoted towards coarse machined ferrous metals and composites, suggesting bead blasted finishes to complement the raw, industrial aesthetic. Confidence scores accurately reflected the clarity of the input; complex, messy sketches generated scores below threshold, triggering the fallback mechanism. This demonstrates that the pipeline accurately reflects the emotional form language theory, correctly correlating visual geometry with abstract emotional categories.

4.3 Cost Sensitivity Analysis

The sensitivity analysis conducted across three material families (Aluminium Alloy, Titanium Alloy, and CFRP) clearly maps the transition points in unit economics. The plotted sensitivity curves demonstrate the profound impact of tooling amortisation. At low production volumes, the unit cost is heavily dominated by the fixed tooling costs, causing the curves for all materials to spike upward. As production volume increases, the

amortised tooling cost approaches zero, and the curves flatten out, dominated by the baseline material and processing costs.

This data is critical for design decision making. By analysing the intersection of these curves, a designer can pinpoint the exact production volume where investing in a more expensive material with cheaper tooling becomes more economical than a cheaper material requiring expensive, high-volume molds. For instance, the curve illustrates the volume threshold at which the high material cost of CFRP is offset by its specific manufacturing processes compared to the extensive machining required for titanium.

4.4 Discussion: How Research Informed the Model

The architecture of MatSel Pro is a direct computational implementation of established aesthetic and emotional design theories. Norman's three-level emotional design model is deeply embedded in the system's logic. The visceral level, concerning immediate visual impact, is operationalised via the Ra and GU scoring metrics in Module 1, treating surface finish as a quantifiable surrogate for visceral appeal. The behavioral level, addressing usability and function, is addressed through the rigorous process compatibility checks and the cost model, ensuring the design is practically executable. The reflective level is captured by the compliance and sustainability scoring, addressing the user's broader values and safety requirements.

Similarly, Hekkert's "most advanced yet acceptable" principle heavily influenced the design of the dynamic weight vectors. In the 'Premium/Luxury' segment, the aesthetic weight is paramount, yet the cost weight is not reduced to zero. This ensures the system recommends advanced, highly appealing materials while maintaining an acceptable economic reality.

The 6S framework (Skeleton, Silhouette, Surface, Seam, Signal, Story) also finds direct mapping within the platform. The Skeleton corresponds to the structural materials evaluated; the Surface is explicitly managed by the metrics; and the Story is inferred and classified by the Image Intelligence module's aesthetic parameter. Our taxonomy of

aesthetic intent (luxury, functional, organic, industrial, playful, minimal) is a direct computational representation of the form languages studied in our literature review. We observed that biomimicry principles directly inform the 'organic' category, driving the recommendation engine to suggest materials with earth tones and softer Ra profiles that align with natural forms. This proves that mechanical engineering specifications and aesthetic theory can be tightly integrated into a single cohesive framework.

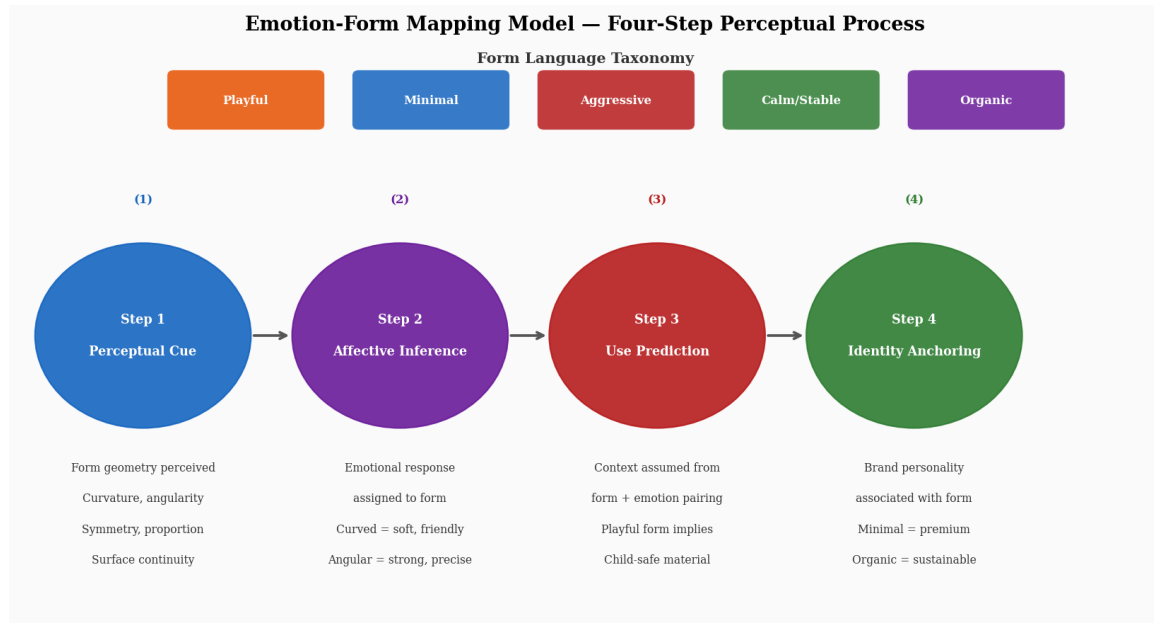


Figure 4.1: Emotion-Form Mapping Model

5. Conclusions and Summary

This project successfully bridged the subjective realm of aesthetics and the objective discipline of mechanical engineering. We conducted extensive research into theories of form generation, emotion-form mapping, and biomimicry, demonstrating how visual perception is rooted in physical geometry. Building upon this theoretical foundation, we developed MatSel Pro, a comprehensive web platform that operationalises these aesthetic principles computationally.

A key finding of this research is that surface roughness (Ra) and gloss units (GU) serve as critical, measurable bridges between rigorous engineering specifications and the user's aesthetic experience. By quantifying these metrics, we established a framework where mechanical engineering and product aesthetics genuinely intersect. Furthermore, the successful deployment of the Image Intelligence module proves that zero-shot vision classification can accurately extract aesthetic intent from preliminary sketches. It effectively maps abstract emotional descriptors to concrete material recommendations without necessitating task-specific model training.

Future work will focus on expanding the material database to include over 10,000 real-world references and incorporating over 100 distinct manufacturing processes. We also plan to integrate an Ashby chart generator for advanced material selection visualisation and implement user-adjustable weight sliders, allowing designers even greater control over the aesthetic and mechanical balancing in the recommendation engine.

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